

Assignment : 3 (CA-Software)

1) Serverless Architecture Benefits :-

Serverless architecture allows developers to build and deploy applications without managing the underlying infrastructure. It offers automatic scaling, reduced operational cost, and pay-per-use billing. Functions execute in response to events, making it ideal for real-time data processing and backend services.

2) Progressive Web Apps (PWAs) :-

PWA, provide a native app-like experience on the web using modern APIs. They work offline, load quickly, and can be installed on devices. PWAs enhance user engagement while reducing development overhead making them suitable for responsive e-commerce or social platforms.

3) AI and Machine Learning in Software Architecture :-

AI and ML enable intelligent behaviour in software, such as recommendation systems, fraud detection and predictive analytics. Architecturally, they are integrated through specialised services or models deployed via containers, APIs, or serverless functions, often requiring data pipelines and model lifecycle management.

* Cloud Computing Service Models :-

1) SaaS (Software as a Service) :-

Provides ready-to-use software (eg, Gmail, Salesforce).

② * PaaS (Platform as a service)
offers a platform for developers to build apps
without managing infrastructure (eg. Heroku, Google
App Engine).

③ IaaS (Infrastructure as a service)
Delivers Virtualized Computing resources (eg. AWS
EC2, Azure VM) for full control over
infrastructure setup.

Each model supports different use cases depending
on control, customization, and scalability needs.